

Can we certify an environment *always safe* by a closed-form
min/max of $c(x)$?

Short answer: no — and the reason is the paper’s thesis

SPARK goal-insertion notes

What “always safe” reduces to

For the **constant-demand** family (SSA, r-SSA, p-SSA) the no-failure certificate is

$$\min_{x \in \mathcal{R}_{\text{active}}} c(x) \geq \eta,$$

and for the **proportional** family (SSS, CBF, r-SSS, r-CBF) it is

$$\lambda \Delta t c_{\max} \leq c_{\min}, \quad c_{\min} = \inf_{\mathcal{R}_{\text{active}}} c(x), \quad c_{\max} = \sup_{\mathcal{R}_{\text{active}}} c(x).$$

So “is this environment always safe?” *is* the question “what are $\min c(x)$ and $\max c(x)$ over the reachable active set?” The instinct to reach for min / max of c is the right object. The obstruction is three independent layers, each fatal to a closed form.

Layer 1 — the domain is the reachable active set (the intractable object)

The certificate quantifies over $x \in \mathcal{R}_{\text{active}}$, the reachable active set — exactly the viability-kernel / Hamilton–Jacobi-reachability object that is subject to the curse of dimensionality past ~ 5 –6 states. If we instead minimize over *all* of configuration space $\mathcal{C}$ we get only a **conservative lower bound**,

$$\min_{x \in \mathcal{C}} c(x) \geq \eta \implies \min_{x \in \mathcal{R}_{\text{active}}} c(x) \geq \eta,$$

which is sound but refuses to certify environments that are actually safe. The *sound* version needs the intractable set; the *tractable* relaxation is loose (and still not closed form — see Layer 2).

Layer 2 — even the global min over $\mathcal{C}$ is nonconvex and non-smooth

Recall

$$c(x) = \sum_k u_{\text{lim},k} \left| [n^\top J(x) G(x)]_k \right|.$$

Three sources of non-closed-form-ness stack:

- $J(x)$ is **trigonometric** in the joint angles $\implies$ nonconvex landscape;
- the $|\cdot|$ and the ℓ_1 sum make c **piecewise-smooth** (kinks);

- the contact normal n is itself an arg min of a nearest-point distance to a nonconvex keep-out, so c is **piecewise-defined** by which obstacle feature is closest.

No formula for the global minimum exists except in degenerate low-DoF / single-obstacle cases. Worse, the infimum is generically pinned by the singularity set: wherever a near-singular contact is reachable,

$$n^\top J(x)G(x) \longrightarrow \mathbf{0} \implies c(x) \longrightarrow 0,$$

so $\min c$ tends to 0 or a tiny value fixed by geometry, not by a clean expression.

Layer 3 — $\min_i c_i(x)$ is not even the right quantity under conflict

With several constraints active at once, feasibility is the **LP value**, not a per-constraint ceiling:

$$\mu(x) = \min_{u \in \mathcal{U}} \max_i (a_i^\top u + \beta_i) \leq 0, \quad a_i := L_g \phi_i^\top, \quad \beta_i := \eta_i + L_f \phi_i,$$

equivalently, over the blend simplex $\Delta = \{y \geq 0, \sum_i y_i = 1\}$,

$$\text{feasible} \iff \forall y \in \Delta: \quad c_y(x) \geq \sum_i y_i d_i, \quad c_y(x) = \sum_k u_{\text{lim},k} \left| \left(\sum_i y_i a_i \right)_k \right|.$$

Checking each $c_i(x) \geq \eta$ only covers the **corners** $y = e_i$: it is *necessary but not sufficient*. A conflicting blend y^* — the **pincer** — can be infeasible,

$$\exists y^* \geq 0: \quad c_{y^*}(x) < \sum_i y_i^* d_i \quad \text{with} \quad \sum_i y_i^* a_i \approx \mathbf{0},$$

while every single-constraint test $c_i(x) \geq \eta$ passes. So a per-constraint min-of- c scalar *misses the dominant failure mode being attacked*; the true test is an LP per state, with no scalar closed form.

What *is* closed form

- **Pointwise** $c(x)$ — evaluating the ruler at a given state is closed form and simulation-free:

$$c(x) = \left\| \text{diag}(u_{\text{lim}}) (n^\top J(x)G(x))^\top \right\|_1.$$

It is the *quantifier* over x that is hard, not the evaluand.

- **The zero set** does have a clean description — a variety:

$$\{c(x) = 0\} = \{x: n(x) \perp \text{Im}(J(x)G(x))\}.$$

- **A conservative upper bound on $c_{\max}$** (since $\|n\|_2 = 1$):

$$c(x) \leq \sum_k u_{\text{lim},k} \|(J(x)G(x))_k\|_2,$$

useful for the overshoot term $\lambda \Delta t c_{\max}$ — but the wrong direction for the constant- η certificate, which needs a *lower* bound on the min.

Why this is a feature: the defender/attacker asymmetry

Certifying “always safe” is a

$$\forall x \in \mathcal{R}_{\text{active}} : \quad g(x) \geq 0 \quad \left(g(x) := \min_{i \text{ active}} c_i(x) - d_i - L_f \phi_i \right)$$

— a universal quantifier over an intractable, nonconvex, multi-constraint set. The defender cannot get it in closed form, so cannot fence the environment off a priori. The **attacker needs only the negation**: a *single* reachable witness

$$\exists x_t : \quad g(x_t) < 0 \quad (\text{or } \mu(x_t) > 0),$$

which is exactly what sampling + admissible-goal fuzzing produces cheaply. The closed-form asymmetry — **no formula to prove universal safety, but a cheap procedure to exhibit one failure** — is precisely why infeasibility is an *attack surface* and not a merely theoretical gap.